Waveshare ESP32-S3-Relay-6CH

Dometic Brisk II Direct Wiring (No Control Box)

SSR-25DA Compressor + Direct Fan Relay Switching

1. Overview

This document describes how to wire the Waveshare ESP32-S3-Relay-6CH module to control the Dometic Brisk II rooftop AC unit. The Dometic Etratech control box (P/N 3313199.000) is eliminated. The Waveshare module has 6 built-in 10A relays and controls all loads via two switching stages:

- SSR-25DA solid state relay for the compressor (25A capacity, handles ~12A running current and inrush). Triggered by CH2 relay via 12VDC.
- Waveshare CH3/CH4 relays for fans (10A contacts, adequate for ~2-3A fan motors). These switch 120VAC directly.
- Waveshare CH1 relay for furnace (dry contact closure to standalone unit with its own blower).

CH2 does NOT switch 120VAC for the compressor. Instead, it switches 12VDC from the RV battery to the SSR-25DA DC input, which then switches 120VAC to the compressor. This keeps all high-current switching on the SSR while CH2 handles only milliamp-level 12VDC trigger current.

WARNING: This system switches 120VAC through the SSR-25DA and Waveshare relay contacts. All 120VAC wiring must be in a proper junction box with 14 AWG wire (minimum for 15A circuit). The SSR-25DA must be mounted on a heatsink. Disconnect shore power before all wiring work.

2. System Architecture

Compressor Signal Path

ESP32 GPIO 2 HIGH -> Waveshare CH2 relay closes -> 12VDC from RV battery reaches SSR-25DA DC+ input -> SSR turns ON -> 120VAC LINE flows through SSR AC output -> Supco SFPC freeze stat (NC) -> 6-pin Blue wire -> Compressor motor

The SSR-25DA handles the full compressor load current (25A rating vs ~12A running). The CH2 relay only switches 12VDC at milliamps to trigger the SSR. This eliminates relay contact wear and welding risk.

Fan Signal Path

ESP32 GPIO 41/42 HIGH -> Waveshare CH3/CH4 relay closes -> 120VAC LINE flows directly through relay NO contact -> 6-pin Red/Black wire -> Fan motor. Fan motors draw ~2-3A, well within the Waveshare 10A contact rating.

Freeze Protection (Dual-Layer)

With the Etratech control box removed, the built-in K1 freeze relay is gone. Freeze protection is now provided by two independent layers:

- SOFTWARE: DS18B20 waterproof probe on the evaporator coil, connected to ESP32 GPIO 10 (via Pico header inside the Waveshare case). When evaporator temp drops below 32F, the ESP32 cuts the compressor (GPIO 2 LOW). Held off until evaporator recovers to 45F.
- HARDWARE BACKUP: Supco SFPC freeze protection control. Clamp-on device that snaps onto the suction line. Normally closed, opens at 35F, closes at 50F. Wired in series between SSR AC2 output and the 6-pin Blue wire. Independent of ESP32 -- passive, unpowered, fail-safe.

3. Component Reference

Waveshare ESP32-S3-Relay-6CH

Spec	Value
MCU	ESP32-S3-WROOM-1U-N16 (16MB Flash, no PSRAM)
Relays	6 channels, 10A @ 250VAC per channel, 1NO+1NC
Trigger	Active HIGH (GPIO HIGH = relay ON)
Power	7-36V DC input or 5V USB-C
Enclosure	DIN-rail ABS case (145 x 90 x 40mm)
Expansion	40-pin Pico-compatible header (inside case)
Onboard	WS2812B RGB LED (GPIO 38), buzzer (GPIO 21)
RS485	GPIO 17 TX, GPIO 18 RX (hardwired to transceiver)

Relay channel assignments:

- CH1 (GPIO 1): Furnace dry contact closure
- CH2 (GPIO 2): Compressor -- switches 12VDC to SSR-25DA DC+ trigger
- CH3 (GPIO 41): Fan Low -- switches 120VAC directly
- CH4 (GPIO 42): Fan High -- switches 120VAC directly
- CH5 (GPIO 45): Expansion (dehumidifier, currently disabled)
- CH6 (GPIO 46): Expansion (vent fan, currently disabled)

SSR-25DA Solid State Relay

Spec	Value
Model	Twtade SSR-25DA
Output	24-380VAC, 25A
Input	3-32VDC (triggered with 12VDC from RV battery)
Heatsink	Required (~15W dissipation at 12A load)

Terminal assignments:

- DC+ (pin 3): 12VDC from Waveshare CH2 relay NO output
- DC- (pin 4): 12VDC GND (RV battery negative)
- AC1 (pin 1): 120VAC LINE (hot) input
- AC2 (pin 2): Output to freeze stat -> 6-pin Blue wire

> Why not trigger the SSR directly from 3.3V GPIO? The SSR-25DA needs ~5mA minimum to trigger reliably. At 3.3V through its internal resistor and LED, only ~1.5mA is available. Using the Waveshare CH2 relay to switch 12VDC provides ~15mA -- well above the minimum threshold.

Supco SFPC Freeze Protection Control

Spec	Value
Model	Supco SFPC
Type	Clamp-on, snaps onto suction line (up to 7/8" OD)
Contacts	SPST NC, opens at 35F, closes at 50F
Rating	10A @ 120V, 5A @ 240V
Install	On suction line just outside evaporator coil

Alternative: Honeywell FPC (opens at 38F, closes at 48F). Either device works. The Supco SFPC has a wider differential (15F) which reduces nuisance trips.

4. Wiring Reference

6-Pin Cable (AC Unit Side)

This cable runs from the junction box up to the Dometic Brisk II rooftop unit. All load wires carry 120VAC when their respective relay/SSR is ON.

Pin	Color	Function	Connection
1	Blue	Compressor	SSR AC2 -> freeze stat -> Blue
2	Black	Fan High	CH4 (GPIO 42) NO -> Black
3	Yellow	Rev Valve	Not connected (no heat pump)
4	Red	Fan Low	CH3 (GPIO 41) NO -> Red
5	White	Neutral	Pass-through to AC unit
6	Grn/Ylw	Chassis Gnd	Pass-through to AC unit

Waveshare Relay Connections

Channel	GPIO	Switches	Connection
CH1	GPIO 1	Dry contact	Furnace call (no voltage)
CH2	GPIO 2	12VDC	SSR DC+ trigger (from RV battery)
CH3	GPIO 41	120VAC	Fan Low (direct to Red)
CH4	GPIO 42	120VAC	Fan High (direct to Black)
CH5	GPIO 45	Reserved	Expansion (dehumidifier)
CH6	GPIO 46	Reserved	Expansion (vent fan)

CH2 (Compressor) is NOT on the 120VAC bus. Its COM connects to RV +12VDC, and its NO connects to SSR DC+. The SSR handles 120VAC switching. CH1 (Furnace) is a dry contact. Only CH3 and CH4 carry 120VAC.

5. Wiring Procedure

DISCONNECT SHORE POWER before all wiring. Verify with a non-contact voltage tester. All 120VAC connections must be inside a junction box.

Step 1: Remove the Etratech Control Box

- Disconnect shore power and verify no voltage is present.
- Disconnect the 6-pin cable from the control box output side.
- Disconnect 120VAC LINE and NEUTRAL from the control box input side.
- Disconnect furnace wires from the control box FURNACE terminals.
- Disconnect freeze sensor wires from J4 (no longer used by control box).
- Remove the Etratech control box entirely.

Step 2: Mount and Wire the SSR-25DA

- Mount the SSR-25DA on a metal heatsink (aluminum, at least 4"x4"). Use thermal paste between SSR and heatsink. Mount heatsink inside or adjacent to the junction box with adequate airflow.
- AC1 (pin 1): Connect with 14 AWG wire to 120VAC LINE (hot). Use a Wago 221-412 or wire nut to branch LINE to both the SSR AC1 and the CH3/CH4 COM bus.
- AC2 (pin 2): Connect with 14 AWG wire to the Supco SFPC freeze stat input terminal. The freeze stat output connects to the 6-pin Blue wire.
- DC+ (pin 3): Connect with 18-22 AWG wire to Waveshare CH2 NO screw terminal.
- DC- (pin 4): Connect with 18-22 AWG wire to RV 12VDC GND (battery negative / chassis ground).

Step 3: Wire 120VAC Bus to CH3/CH4 COM

- Create a 120VAC LINE bus to CH3 COM and CH4 COM screw terminals. Use 14 AWG wire and Wago connectors or wire nuts to branch from the same LINE wire feeding SSR AC1.
- CH3 NO (GPIO 41) -> 14 AWG -> 6-pin Red wire (Fan Low)
- CH4 NO (GPIO 42) -> 14 AWG -> 6-pin Black wire (Fan High)

Step 4: Wire 12VDC to CH2 COM

- Connect RV +12VDC (house battery / converter) to CH2 COM screw terminal. Use 18-22 AWG wire. This is the SSR trigger source.
- CH2 NO connects to SSR DC+ (already done in Step 2).
- When GPIO 2 goes HIGH, CH2 closes, 12VDC reaches SSR DC+, and the SSR turns on 120VAC to the compressor.

Step 5: Wire Neutral and Ground Pass-Through

- Route NEUTRAL (white) directly through the junction box to the 6-pin White wire. Not switched by any relay.
- Route GROUND (green/yellow) directly through to the 6-pin Green/Yellow wire. Connect ground to the junction box and any metal enclosures.

- Cap the Yellow wire (Pin 3, reversing valve) with a wire nut.

Step 6: Wire Furnace (CH1)

- Connect furnace call wires to CH1 COM and NO screw terminals. This is a dry contact closure -- no 120VAC, no 12VDC on these terminals.

Step 7: Install Supco SFPC Freeze Stat

- Snap the Supco SFPC clamp onto the suction line just outside the evaporator coil. No thermal paste needed -- the clamp provides direct contact with the copper tubing.
- Wire in series: SSR AC2 output -> freeze stat -> 6-pin Blue wire. Use 14 AWG wire and appropriate connectors.
- Verify NC (closed) at room temperature with a continuity test.

Step 8: Install DS18B20 Freeze Sensor

- Mount the waterproof DS18B20 probe on the evaporator coil in contact with copper tubing or aluminum fins. Use thermal paste and secure with cable tie or aluminum tape.
- Open the Waveshare case to access the 40-pin Pico expansion header. Wire: Red to 3.3V (Pin 36), Black to GND (Pin 3/8/13), Yellow/White data to GPIO 10.
- Solder a 4.7K ohm pull-up resistor between GPIO 10 and 3.3V, as close to the Pico header as practical.
- Route sensor cable away from 120VAC wires to avoid noise.

6. I2C Sensor and Display Wiring

The BME280 temperature/humidity/pressure sensor and SSD1306 OLED display are mounted at the thermostat location (former Dometic stat position) and connected to the Waveshare module via a 3-meter cable run through the Pico expansion header inside the case.

Pico Header Pin Connections

Signal	GPIO	Pico Pin	Wire Color
SDA	GPIO 8	Pin 32 (GP27)	Green
SCL	GPIO 9	Pin 34 (GP28)	Blue
3.3V	--	Pin 36 (3V3)	Red
GND	--	Pin 3/8/13	Black

I2C Devices (Shared Bus)

Device	Address	Function
BME280	0x76	Temperature, humidity, pressure
SSD1306 OLED	0x3C	128x64 pixel display

Both devices share the same I2C bus (SDA + SCL). Use 18 AWG 4-conductor thermostat wire for the 3-meter cable run. I2C bus speed: 400 kHz (can reduce to 100 kHz if cable causes issues). WARNING: OLED and BME280 VCC = 3.3V ONLY (not 5V or 12V!).

7. Freeze Protection Details

Software Layer (DS18B20)

Condition	Action
Evap temp < 32F	Cut compressor (GPIO 2 LOW)
Evap temp >= 45F	Allow compressor to resume
DS18B20 not found	Software freeze protection disabled
Check interval	Every 5 seconds

Hardware Layer (Supco SFPC)

Condition	Action
Suction line < 35F	Switch opens, compressor circuit broken
Suction line > 50F	Switch closes, circuit restored
No power needed	Passive NC switch, always active

The two layers are independent. The DS18B20 software layer acts first (32F threshold). The Supco SFPC hardware layer is backup (35F threshold). Even with a total ESP32 failure, the SFPC physically breaks the compressor circuit if the suction line reaches 35F.

8. Failsafe Behavior

The system is fail-safe: any single point of failure results in HVAC OFF. The Waveshare relays and SSR-25DA are normally open, so power loss to the ESP32 disconnects all loads.

Condition	Result
ESP32 power loss	All relays open -> SSR off -> all loads OFF
ESP32 software crash	boot.py sets GPIOs LOW on reboot -> all OFF
Shore power loss	No 120VAC available -> all loads OFF
12VDC battery loss	SSR DC+ has no trigger -> compressor OFF
DS18B20 reads < 32F	Software cuts compressor, recovery at 45F
SFPC trips (< 35F)	Hardware breaks compressor circuit
SSR failure (open)	Compressor stays OFF (safe direction)
Relay welded shut	Only that one fan load stays on
120VAC bus fault	Breaker trips -> all loads OFF

There is NO backup thermostat. If the ESP32 loses power or crashes, all HVAC stops until the ESP32 recovers. The system is fail-safe (everything OFF) but not fail-operational.

9. Waveshare ESP32-S3-Relay-6CH GPIO Reference

IP: 192.168.71.153 | DIN-rail enclosure | 7-36V DC or 5V USB-C

GPIO	Function	Channel/Device	Load
1	Furnace	CH1 NO/COM	Dry contact closure
2	Compressor	CH2 -> SSR	12VDC trigger -> SSR 120VAC
41	Fan Low	CH3 NO	120VAC -> Pin 4 Red
42	Fan High	CH4 NO	120VAC -> Pin 2 Black
45	Expansion	CH5	Dehumidifier (disabled)
46	Expansion	CH6	Vent fan (disabled)
8	I2C SDA	Pico Pin 32	BME280 + OLED
9	I2C SCL	Pico Pin 34	BME280 + OLED
10	Freeze Sensor	Pico header	1-Wire DS18B20, 4.7K pull-up
38	RGB LED	Onboard	WS2812B status indicator
21	Buzzer	Onboard	Passive buzzer for alerts
17	RS485 TX	Onboard	Hardwired to transceiver
18	RS485 RX	Onboard	Hardwired to transceiver

Power Architecture

- Control side: RV 12VDC battery system (always available with battery). Powers Waveshare module (7-36V DC input) and SSR trigger via CH2.
- Load side: 120VAC shore power through SSR and relay contacts. Completely isolated from control side via relay contacts.
- boot.py sets CH1-CH4 GPIOs (1, 2, 41, 42) LOW (OFF) on startup.

10. Pre-Installation Checklist

Tools Required

- Digital multimeter with continuity and AC voltage modes
- Non-contact voltage tester (NCVT)
- Wire strippers rated for 14 AWG and 18-22 AWG
- Crimping tool for ring/spade terminals (if used)
- Small screwdriver for Waveshare screw terminals and SSR terminals
- Soldering iron (for DS18B20 pull-up resistor)
- Heat shrink tubing and heat gun
- Cable ties and/or aluminum tape (for sensor mounting)

Materials

- Waveshare ESP32-S3-Relay-6CH module (DIN-rail)
- Twtade SSR-25DA solid state relay
- Aluminum heatsink for SSR (at least 4"x4")
- Thermal paste (for SSR-to-heatsink and DS18B20 mounting)
- Supco SFPC freeze protection control (clamp-on)
- DS18B20 waterproof temperature sensor (stainless steel probe)
- 4.7K ohm resistor (1/4W, for DS18B20 pull-up)
- BME280 sensor breakout board
- SSD1306 OLED display (128x64, I2C)
- 18 AWG 4-conductor thermostat wire (~3 meters for I2C run)
- 14 AWG stranded copper wire (THHN or equivalent), 3-4 feet
- 18-22 AWG wire for 12VDC SSR trigger connections
- Wire nuts (14 AWG rated) or Wago 221-412/415 connectors
- Junction box (metal or PVC, sized for conductor count)
- Cable clamps / strain reliefs for junction box knockouts

Verification Checklist

- [ ] Shore power DISCONNECTED and verified with NCVT
- [ ] Etratech control box fully removed from circuit
- [ ] SSR-25DA mounted on heatsink with thermal paste
- [ ] SSR AC1 (pin 1) connected to 120VAC LINE (14 AWG)
- [ ] SSR AC2 (pin 2) -> Supco SFPC -> 6-pin Blue (compressor)
- [ ] SSR DC+ (pin 3) connected to Waveshare CH2 NO terminal
- [ ] SSR DC- (pin 4) connected to RV 12VDC GND
- [ ] CH2 COM connected to RV +12VDC (NOT 120VAC)
- [ ] 120VAC LINE bus connected to CH3/CH4 COM only (14 AWG)

- [ ] CH3 NO (GPIO 41) -> 6-pin Red (fan low)
- [ ] CH4 NO (GPIO 42) -> 6-pin Black (fan high)
- [ ] CH1 (GPIO 1) wired as dry contact for furnace ONLY
- [ ] CH1 NOT connected to 120VAC or 12VDC bus
- [ ] Neutral (White) passes through directly -- NOT switched
- [ ] Ground (Green/Yellow) passes through directly -- NOT switched
- [ ] Yellow wire (reversing valve) capped with wire nut
- [ ] Supco SFPC clamped on suction line near evaporator
- [ ] Supco SFPC verified NC (closed) at room temperature
- [ ] DS18B20 wired to GPIO 10 (Pico header) with 4.7K pull-up to 3.3V
- [ ] DS18B20 probe mounted on evaporator coil with thermal paste
- [ ] I2C cable: SDA=GPIO 8 (Pin 32), SCL=GPIO 9 (Pin 34), 3.3V, GND
- [ ] BME280 and OLED responding on I2C bus
- [ ] All 120VAC connections inside junction box
- [ ] All wire nuts / Wago connectors tight and secure
- [ ] Junction box closed and cable clamps installed
- [ ] Shore power reconnected
- [ ] ESP32 boot verified: all relays OFF on startup
- [ ] Each relay tested individually from web UI
- [ ] SSR trigger verified: GPIO 2 HIGH -> compressor runs
- [ ] DS18B20 reading verified in web UI (reasonable temperature)
- [ ] Freeze protection tested: sensor below 32F -> compressor cuts

Generated for RV Thermostat Project - ESP32 MicroPython